ℓ ALONG THE $-1/n$ SURGERIES ON A FIBRE OF A BRIESKORN SPHERE

Labels as in `status.pdf`. Records: `notes/R6/brieskorn.md`, code and tables in `notes/R6/brieskorn-code/`. An independent recheck is running.

1. Question and conventions

$\Sigma = \Sigma(a_1, a_2, a_3)$ with its link orientation (boundary of the negative definite plumbing); K the singular fibre of order a_K; P the product of the other two orders. The -1 slope is the homology-sphere slope whose surgery cobordism is negative definite (-1 on the right-handed trefoil gives $\Sigma(2,3,7)$). Levels are $c = (3+\rho)/16$; on Brieskorn spheres the differential vanishes, so $\ell = \min c$. Question: do $\ell(Y_{-1}), \ell(Y_{-1/2}), \ell(Y_{-1/3})$ decrease, and is $T_1 = \ell(Y_{-1}) - \ell(Y_{-1/3}) \geq \frac{1}{8}$?
Topology (proved; checked by Casson's surgery formula on every manifold). $\Sigma_{-1/n}(K) = \Sigma(\dots, a_K + nP, \dots)$ and $(-\Sigma)_{-1/n}(K) = \operatorname{sgn}(nP - a_K)\,\Sigma(\dots, |nP - a_K|, \dots)$. More invariantly: the homology-sphere fillings of the exterior of K form one chain $Z_k = \operatorname{sgn}(r + kP)\,\Sigma(a_j, a_k, |r + kP|)$, $r \equiv a_K \bmod P$, each obtained from the previous one by -1 surgery on the core. Z_k is link-oriented for $k \geq 0$ and reversed for $k \leq -1$. Torus knots in S^3 are the chains with $r = 1$ or $r = P - 1$; the base S^3 sits at Z_0 or Z_{-1}.

2. Answers (computed)

Range: all 3122 triples with $a_1 a_2 a_3 \leq 5000$ that are not torus-knot surgeries (18,732 families: each fibre, both orientations), 3036 torus-knot families, and all 3300 exteriors with $P \leq 120$ followed along $Z_{-6}, \dots, Z_{10}$ (77 of them to Z_{40}).

base	decreasing	$\min T_1$	$T_1 = \frac{1}{8}$
Σ (9366 families)	always	$\frac{1}{8}$	77
$-\Sigma$, $a_K < P$ (7739)	always	$\frac{1}{8}$	539
$-\Sigma$, $a_K > P$ (1627)	fails for $2P < a_K < 4P$	$-\frac{9}{8}$	0

So: yes on the link-oriented side, with $\frac{1}{8}$ sharp. In the first two rows every $Y_{-1/n}$ is link-oriented. In the third row the first surgeries are reversed: $T_1 \geq \frac{1}{8}$ exactly when $a_K < 3P$; for $3P < a_K < 4P$, T_1 is mostly ≤ 0; for $a_K > 4P$, $0 < T_1 < \frac{1}{8}$ and $T_1 \to 0$. First failure: $K = 23 \subset -\Sigma(2,5,23)$, where $Y_{-1}, Y_{-1/2}, Y_{-1/3} = -\Sigma(2,5,13), -\Sigma(2,3,5), \Sigma(2,5,7)$ have $\ell = \frac{209}{520} < \frac{59}{120} > \frac{11}{70}$. Along every chain ℓ goes down at each step except $Z_{-2} \to Z_{-1}$, which goes up in 3032 of 3090 non-torus chains: Z_{-1} occupies the slot that S^3 ($\ell = \infty$) occupies in a torus-knot chain. Negative definiteness of the cobordisms alone therefore does not give the $\frac{1}{8}$.
Equality. $T_1 = \frac{1}{8}$ exactly when the minimisers at both ends have traceless holonomy along the surgered fibre (5067 equality windows, none of the 75,828 others). Examples: $K = 17 \subset \Sigma(3,5,17)$; $\ell(\Sigma(p,q,2+2jpq)) = \ell(\Sigma(2,p,q)) - \frac{j}{8}$ for odd coprime p, q. This is the mechanism already seen for odd–odd mirror torus knots: a sphere of square -2 in the two-step cobordism, with boundary $\mathbb{R}P^3$ and the rotation by π there, contributing $\frac{1}{8}$.

3. The exact form (conjecture)

On the link-oriented part of a chain, with m the order of the surgered fibre and $\alpha \in [0, \frac{1}{2})$ the holonomy of the ℓ-minimiser along it (eigenvalue $e^{\pm 2\pi i\alpha}$, so $\alpha = \frac{1}{4}$ is traceless),

$$\ell\big(\Sigma(a_j, a_k, m)\big) = A^* - \frac{m}{P}\,\alpha\big(\tfrac{1}{2} - \alpha\big), \qquad A^* \text{ constant on the chain.}$$

Writing $\beta = 4m|\alpha - \frac{1}{4}|$ (an integer, constant on each parity class of the chain), this is $\ell = A^* - (m^2 - \beta^2)/(16Pm)$. It holds exactly in 80,894 of 80,895 windows; in the remaining one a tied pair of generators undercuts it and $T = 0.141$. For $T(p,q) \subset S^3$: $\ell(S^3_{-1/n}) = \frac{3}{8} - \frac{5}{16pq} - \frac{n}{16} + \frac{\beta_n^2}{16pq(pqn+1)}$ with $\beta_n = 3$ (n even), $pq - 3$ (n odd). The identification of β with the holonomy is checked on 14 parity classes. Consequences:

$$\ell(Y_{-1/n}) - \ell(Y_{-1/(n+2)}) = \frac{1}{8} + \frac{\beta^2}{8\,m_n m_{n+2}}, \qquad \ell(Y_{-1/n}) - \ell(Y_{-1/(n+1)}) \geq \frac{m_n}{16\,m_{n+1}}.$$

Remark. This is the slope term of the integral-surgery level formula, $-N\alpha(\frac{1}{2} - \alpha) = -\frac{N}{16} + N(\alpha - \frac{1}{4})^2$, with $N = m/P$. Each step adds 1 to N. So the asymptotic drop per step is $\frac{1}{16}$, its value at traceless holonomy, and the excess is a perfect square that vanishes exactly there. At generator level the same holds: there is an injection φ from the generators of Z_k to those of Z_{k+2} with $c(\varphi x) = c(x) - \frac{1}{8} - \beta(x)^2/(8m_k m_{k+2})$ in every case tested; so the two-step bound holds generator by generator.
One step. No positive universal bound: $\ell(Y_{-1}) - \ell(Y_{-1/2})$ reaches $\frac{1}{8(P+2)}$ (e.g. $\frac{1}{568}$) for $-\Sigma$ with $a_K < P$. For a link-oriented base every one-step drop exceeds $\frac{1}{24}$ (minimum seen $\frac{625}{14998}$, $K = 2 \subset \Sigma(2,17,147)$).

Reversed side. There $\ell(Z_k) = A' - \gamma^2/(4P|m_k|)$: ℓ is governed by the top of the spectrum of Σ, which is bounded. There is no term linear in k, so two-step drops are $\gamma^2/(2mm'') \to 0$, with no $\frac{1}{8}$.

4. **Conjectures suggested**

1 (two-step). If $Y_{-1/(n+2)}$ is link-oriented, then $\ell(Y_{-1/n}) - \ell(Y_{-1/(n+2)}) \geq \frac{1}{8}$, with equality iff the minimisers are traceless along the surgered fibre. Equality forces P odd, m_n even, and $m_n \bmod 2P$ in a set of residues that always contains 2 and $P-1$.
2 (exact form). The displayed formula, with A^* constant on the chain.
3 (one step). $\ell(Y_{-1/n}) - \ell(Y_{-1/(n+1)}) \geq m_n/(16m_{n+1})$ on the link-oriented side.
What is open. For a knot in a general homology sphere there is no evident replacement for "link-oriented". For knots in S^3 all $-1/n$ surgeries lie on the correct side, and Conjecture 1 is consistent with all torus-knot data. The two-step drop has the shape sought for ± 2: a fixed term from a rotation-by-π end, plus a square. Here, however, it is realised by an explicit continuation of generators along a fixed exterior, which the general case lacks.